

Brazilian Social Media Anti-vaccine Information Disorder Dataset - Telegram (2020-2025)

Overview

The dataset contains about four million Telegram posts collected from 119 Brazilian anti-vaccine channels between January 2020 and June 2025. It was created to support the scientific and public health community in analyzing the spread and evolution of false or misleading information regarding vaccination in Brazil.

The collection captures the period corresponding to a significant decline in national vaccination coverage and the concurrent "infodemic" circulating on digital platforms.

- **Total Messages:** 3,998,633
- **Time Period:** January 2020 – June 2025
- **Total Data Volume:** 5.5 TB (including media)
- **License:** Creative Commons BY-NC-SA 4.0
- **DOI:** [10.25824/edu/5.JIVDT](https://doi.org/10.25824/edu/5.JIVDT)

Data Availability

The text dataset and metadata are openly available for non-commercial research.

- **Text Data:** Available in .jsonl (JSON Lines) format.
- **Media Data:** Access to associated media (images, videos, audio) requires signing a specific agreement outlining ethical responsibilities due to the sensitivity and size of the files.

Dataset Structure

The dataset is formatted in JSON Lines (.jsonl). Each line represents a single message object with the following schema:

Field Name	Type	Description
message_id	String	Unique identifier for the message within its chat.
channel_id	String	Unique identifier for the source channel or group.
user_id	String	SHA-256 hash of the author's user ID (anonymized).
date	Datetime	UTC timestamp indicating when the message was sent.
collected_date	Datetime	UTC timestamp indicating when the message was collected.
edit_date	Datetime	UTC timestamp of the message's last edit.

Field Name	Type	Description
text_content	String	Raw text of the message. Empty for non-text messages.
reply_to_id	String	Identifier of the parent message if this message is a reply.
n_forwards	Integer	Number of times the message has been forwarded.
views	Integer	Number of times the message has been viewed.
reactions	Integer	Total number of reactions received, regardless of type.
forward_from	String	Source channel if the message is a forward.
forward_from_views	Integer	View count of the original forwarded message.
forward_from_reaction	Integer	Reaction count of the original forwarded message.
forward_from_n_forwards	Integer	Forward count of the original forwarded message.
media_type	String	Type of media attached (e.g., document, image, video).
media_url	String	URL associated with the media, if present.
media_title	String	Title of the media, if provided.
media_description	String	Description of the media (as provided by Telegram).
is_vaccine_related	Boolean	Indicates whether the message concerns vaccine-related topics.
language	String	Predominant language of the message (e.g., Portuguese).

Statistics

General Statistics

- **Total Channels/Groups:** 119
- **Unique Anonymized Users:** 71,672
- **Messages with Text:** 3,345,088 (83.6%)
- **Vaccine-Related Posts:** 407,723 (10.2%)
- **Main Languages:** Portuguese (58.3%), English (8.0%), Spanish (1.7%)

Media Distribution

The dataset encompasses 1,440,498 items associated with media content.

Media Type	Count
Image	765,750
Video	645,949
Audio	17,376
Poll	1,931
Others	9,490

Methodology

Collection

Data was collected using a custom Python tool built on the **Telethon** library. The target channels were identified using seed lists from prior literature and keyword searches including terms like "Vacina," "mRNA," "Nova Ordem Mundial," and "Efeitos adversos". **Only public channels with at least 1,000 members were monitored.**

Annotation & Processing

- **Language Detection:** Performed using langdetect with a confidence threshold of 0.5.
- **Topic Classification:** The field is_vaccine_related was generated using the **Sabiá-3** Large Language Model. The model achieved a 90% F1-score against human annotators.
 - *Criteria:* Mentions of vaccines/immunization, efficacy/safety discussions, policy discussions, conspiracy theories, or hesitancy.

Privacy & Anonymization

To comply with ethical guidelines and Brazilian data protection laws (LGPD):

- **User/Channel IDs:** Replaced with a persistent, non-reversible SHA-256 hash salted with a private key.
- **PII Removal:** Phone numbers, emails, and addresses were redacted from text content using Microsoft Presidio.
- **Join/Leave Events:** All timestamped entries for users joining or leaving channels were removed to prevent re-identification.

Limitations

1. **Engagement Metrics:** The "reactions" feature was only implemented by Telegram in late 2021; data prior to Dec 30, 2021, lacks this metadata.
2. **File Size:** Media files larger than 50MB were excluded from collection.
3. **Deleted Content:** Content removed by channel admins or the Brazilian Supreme Court during the collection period may be missing or altered.